

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2016-17

Paper No. STA/STB-501: Applied Statistics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

- (a) Discuss GRR and NRR in detail with suitable examples. State the interpretation of the statement " $\text{NRR} = 1$ ". [7]
 - (b) Describe the general fertility rate and total fertility rate with examples. Write the interpretation of $\text{TFR} = 2$. [7]
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Question 2.

- (a) What are the methods of obtaining Vital Statistics? What are the various sources of Vital Statistics? Also give the uses of Vital Statistics. [7]
 - (b) Give a brief history of the Census of India. Who was the Registrar General for Census 2011? [7]
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Question 3.

- (a) What do you mean by a life table? Write the assumptions of the life table. Also discuss the various columns of a complete life table. [7]
 - (b) Discuss all the important curves used for measuring the growth of a population (e.g., exponential, logistic). [7]
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Question 4.

- (a) Discuss the importance and limitations of index numbers. [7]
 - (b) Define crude death rate. How does standardisation help in overcoming the limitation of the crude death rate? Explain clearly the standardised death rates, including the methods of their computation. [7]
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Question 5.

- (a) What is a price index number? Explain the meaning of the current year and base year in reference to an index number. [7]
- (b) What is meant by a cost of living index number? Describe the procedure followed in its preparation. [7]
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Question 6.

- (a) Discuss the various tests of index numbers and verify that Fisher's index number is an ideal index number. [7]
- (b) What do you mean by time series? What are its uses? [7]
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Question 7.

- (a) Write short notes on the different types of components of time series. Explain the difference between seasonal and cyclical variations with an example. [7]
- (b) Explain secular trend in detail. Critically examine the various methods of measuring trend. [7]
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Question 8.

- (a) What is statistical quality control (SQC)? Discuss the need and utility of SQC in industry. Explain why 3σ limits are used in SQC charts. [7]
- (b) Discuss in detail the construction of the p -chart and c -chart. What is the theoretical justification for using the Poisson distribution in the case of the c -chart? [7]
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Question 9.

Discuss the construction of the $(\bar{X}, R)$ -chart, explaining the role of control limits and the rationale for using the range R as a measure of variability. [14]
